

Chapter 1 Overview and Descriptive Statistic

1.1 Populations, Samples and Processes

Population

The set of all objects

Variables(Characteristics)

Objects whose value may change from one object to another

According to **the number of variables**, we have

- Univariate: a single variable(e.g. automatic or manual for cars)
- Bivariate: two variables(e.g. the height and weight of the students)
- Multivariate: more than two variables.

Sample

A **subset** of population.

Branches of statistics

- Descriptive statistics:
Summarize and describe important features of data.
 - Graphical methods
 - Numerical summary measures(means, standard deviations, correlations etc.)
- Inferential statistics
Use sample information to draw some type of conclusion about the population. Generalize information **from a sample to a population**.

What is the relation between probability and statistics

Probability reasons **from the population to the sample**.

Inferential statistics reasons **from the sample to the population**.

Collecting Data

- Methods for collecting data
 - Random sampling: each element of population has an **equal chance** to be selected.

- Stratified(by layers) sampling: divide into subpopulation and random samples are taken of each stratum.

1.2 Pictorial and Tabular Method in Descriptive Statistics

Notation

Sample size: The number of observations in a single sample will often be denoted by n .

Stem-and-Leaf Displays

Suppose we have a numerical data set $x_1, x_2, \dots, x_n$ for which x_i consists of at least two digits.

Steps:

- Select one or more leading digits for the stem values, The trailing digits become the leaves.
- List possible stem values in a vertical column(from small to large).
- Record the leaf for every observation beside the corresponding stem value.
- Indicate the units for stems and leaves someplace in the display

Example: 16, 23, 64, 37, 31

Note for indicate the units that stem: ten's digit; leaf: one's digit.

- Identification of a typical or representative value
- Extent of spread
- Presence of any gaps
- Extent of symmetry
- Number and location of peaks
- Presence of any outlying values

Dotplot

The data set is reasonably small or there are relatively few distinct data values

- If repeated value, pile it in the vertical column

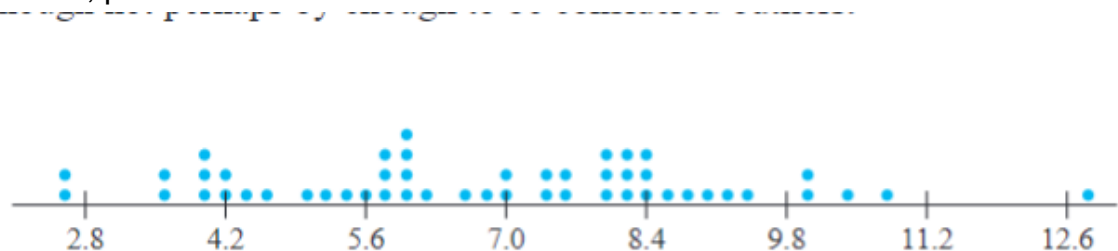


Figure 1.6 A dotplot of the data from Example 1.8

Histogram

Type of variables:

- Discrete variable:

Either is finite or else can be listed in an infinite sequence

Three Steps:

- Determine the frequency(e.g. 70) (or relative frequency(e.g. 70/200)) of each x value
- Make possible x values on a horizontal scale
- Draw a rectangle whose height is the (relative) frequency of the value.
relative frequency of a value equals number of times the value occurs over the number of observations in the sample

- Continuous variable:

Consist of an entire interval

Similar to Discrete Histogram, height of rectangle represents frequency

- Typical Histogram shapes
 - Symmetric unimodal
 - Bimodal
 - Positively Skewed
 - Negatively Skewed

1.3 Measures of Location

The Mean

- Sample Mean:

The sample mean of observations $x_1, \dots, x_n$ is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{(\sum x_i)}{n}$$

- Sample median:

The sample media is obtained by first ordering the n observations from smallest to largest.
(denoted by $\tilde{x}$)

Three different sharps for a population distribution

$\bar{u} = \tilde{u}$: symmetric unimodal

If median is smaller, positively skewed

If median is larger, negative skewed

Other Measures of Location

- Quartiles:

divide the population into four equal partitions(first quartile, second quartile or median, third

quartile)

- Percentile
- Trimmed Means

A compromise between sample mean and sample median. Denoted by $\bar{x}_{tr(n)}$

By eliminating the smallest $n\%$ and the largest $n\%$ and then averaging the left values. (the number of eliminated values **takes flooring**) (general proportion: 5% – 25%)

1.4 Measures of Variability

The range

The difference between the largest and the smallest sample values

Deviations from the mean

$$\sum_{i=1}^n (x_i - \bar{x})^2$$

Sample variance

denoted by s^2

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{S_{xx}}{n - 1}$$

Note it is $n - 1$.

The **sample standard deviation** denoted by s is $s = \sqrt{s^2}$

- An alter expression for the numerator of s^2

$$S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

- If $y_i = x_i + c$, then $s_y^2 = s_x^2$
- if $y_i = cx_i$, then $s_y^2 = c^2 s_x^2$, $s_y = |c| s_x$

Population variance

σ^2 to denote the population variance and σ to denote the population standard deviation.

$$\sigma^2 = \sum_{i=1}^N (x_i - \mu)^2 / N$$

where μ is the means of population.

Boxplots

- Fourth spread f_s

f_s = upper fourth quartile - lower fourth quartile.

The median is included in both halves if n is odd.

The simplest boxplot is based on the **5-number summary**: smallest, lower fourth, median, upper fourth, largest

40 52 55 60 70 75 85 85 90 90 92 94 94 95 98 100 115 125 125

The five-number summary is as follows:

smallest $x_i = 40$ lower fourth = 72.5 $\tilde{x} = 90$ upper fourth = 96.5
largest $x_i = 125$

Figure 1.20 shows the resulting boxplot. The right edge of the box is much closer to the median than is the left edge, indicating a very substantial skew in the middle half of the data. The box width (f_s) is also reasonably large relative to the range of the data (distance between the tips of the whiskers).

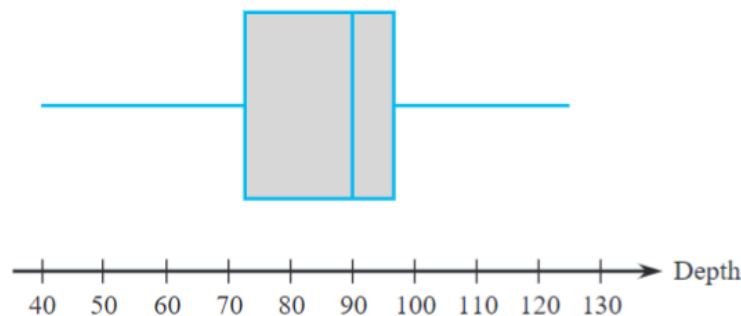


Figure 1.20 A boxplot of the corrosion data

- Outlier: Any observation farther than $15f_s$ from the closest fourth is an outlier
- Extreme: more than f_s
- Mild: $(15f_s, f_s)$

- Comparative Boxplots

